

Mingoo Seok

其他

基本信息

姓名: Mingoo Seok

所属院校: 哥伦比亚大学

地区: United States

学科领域

电子与计算机工程

集成电路设计与集成系统

研究领域

低功耗设计与能耗优化

计算机体系结构

低功耗电路与能量管理

数字集成电路设计

嵌入式系统与硬件设计

深度学习与神经网络

存储器设计与架构

个人简介

Mingoo Seok is a distinguished professor in Electrical Engineering at Columbia University. His research focuses on integrated circuits, VLSI, ultra-low power systems, power management, and machine learning hardware. Notably, he has contributed to the development of XNOR-SRAM and C3SRAM, pioneering SRAM macros for deep neural networks. With an h-index of 41 and over 6,800 citations, his work significantly impacts the field of solid-state circuits and low-power electronics.

工作经历

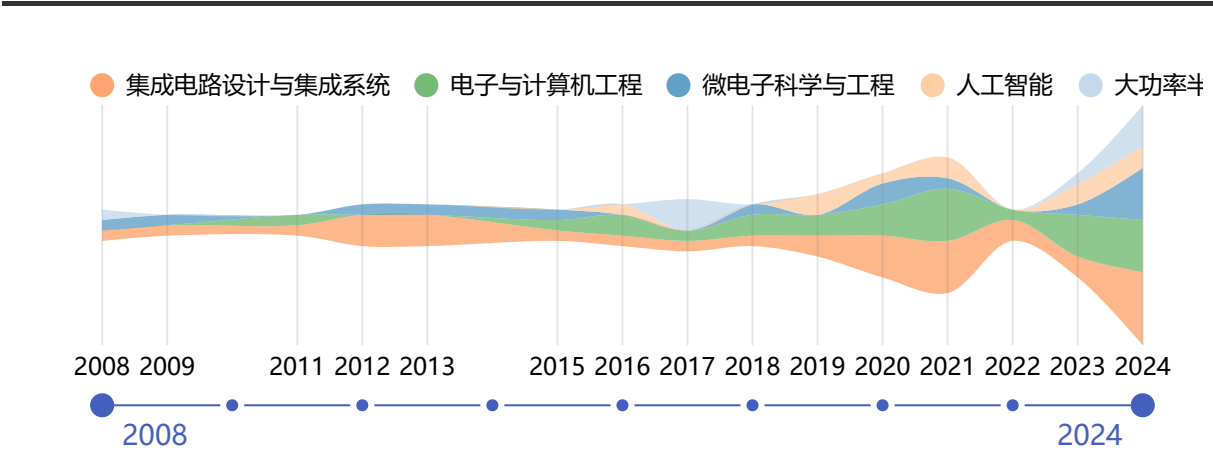
Columbia University

University of Michigan - Professor

Cornell Tech

Columbia University	-
Huawei Technologies, Co., Ltd.	-
Syntiant	-
Apple Inc.	-
Samsung Advanced Institute of Technology	-
Apple	-
Columbia University - Professor	-
University of Michigan	-
University of Michigan	-
Mythic	-
Rutgers University	-
Seoul National University	-
Sungkyunkwan University - Associate Professor	-
Meta Platforms, Inc.	-
Arizona State University - Professor	-
Columbia University - Professor	-
Columbia University - Associate Professor	-

近年发表的论文



A 4.2-to-0.5-V, 0.8- μ A-0.8-mA, Power-Efficient Three-Level SIMO Buck Converter for a Quad-Voltage RISC-V Microprocessor

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Dongkwun; Wang, Zhaoqing; Huang, Paul Xuanyuanliang; Chundi, Pavan Kumar; Kim, Suhwan; Blanco, Andres A.; Krishnamurthy, Ram K.; Seok, Mingoo

被引用次数: 0

2024

MACC-SRAM: A Multistep Accumulation Capacitor-Coupling In-Memory Computing SRAM Macro for Deep Convolutional Neural Networks

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Bo; Saikia, Jyotishman; Meng, Jian; Wang, Dewei; Kwon, Soonwan; Myung, Sungmeen; Kim, Hyunsoo; Kim, Sang Joon; Seo, Jae-Sun; Seok, Mingoo

被引用次数: 1

2024

A Metastability Risk Prediction and Mitigation Technique for Clock-Domain Crossing With Single-Stage Synchronizer in Near-Threshold-Voltage Multivoltage/Frequency-Domain Network-on-Chip

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Chuxiong; He, Weifeng; Sun, Yanan; Shao, Lin; Zhang, Bo; Yang, Jun; Seok, Mingoo

被引用次数: 0

2024

iMCU: A 28-nm Digital In-Memory Computing-Based Microcontroller Unit for TinyML

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Chuan-Tung; Huang, Paul Xuanyuanliang; Oh, Jonghyun; Wang, Dewei; Seok, Mingoo

被引用次数: 2

2024

INTIACC: A Programmable Floating-Point Accelerator for Partial Differential Equations

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Huang, Paul Xuanyuanliang; Tsvidis, Yannis; Seok, Mingoo

被引用次数: 2

2024

A Ten-Level Series-Capacitor 24-to-1-V DC-DC Converter With Fast In Situ Efficiency Tracking, Power-FET Code Roaming, and Switch Node Power Rail

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wang, Zhaoqing; Li, Mao; Kim, Suhwan; Desai, Nachiket; Krishnamurthy, Ram K.; Zhang, Xin; Seok, Mingoo

被引用次数: 0

2024

EQZ-LDO: A Secure Digital Low Dropout Regulator Armed With Detection-Driven Protection Against Correlation Power Analysis

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Mao; Park, Jaehan; Kim, Sung Justin; Kim, Dongkwun; Mathew, Sanu K.; De, Vivek; Seok, Mingoo

被引用次数: 0

2024

A 394-TOPS/W Matched Filter With Charge-Domain Computing for GPS Signal Acquisition

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Jieyu; He, Weifeng; Zhang, Bo; Lin, Chuxiong; Qi, Liang; Liu, Dingxuan; Seok, Mingoo

被引用次数: 0

2024

Background Noise and Process-Variation-Tolerant Sub-Microwatt Keyword Spotting Hardware Featuring Spike-Domain Division-Based Energy Normalization

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Wang, Dewei; Kim, Sung Justin; Yang, Minhao; Lazar, Aurel A.; Seok, Mingoo

被引用次数: 0

2024

A 44.2-TOPS/W CNN Processor With Variation-Tolerant Analog Datapath and Variation Compensating Circuit

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Seo, Jin-O; Seok, Mingoo; Cho, Seonghwan

被引用次数: 1

2024

DIMCA: An Area-Efficient Digital In-Memory Computing Macro Featuring Approximate Arithmetic Hardware in 28 nm

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Chuan-Tung; Wang, Dewei; Zhang, Bo; Chen, Gregory K.; Knag, Phil C.; Krishnamurthy, Ram Kumar; Seok, Mingoo

被引用次数: 6

2024

A Capacitorless External-Clock-Free Fully Synthesizable Digital LDO With Time-Based Load-State Decision and Asynchronous Recovery

IEEE TRANSACTIONS ON POWER ELECTRONICS

Oh, Jonghyun; Song, Yoonho; Hwang, Young-Ha; Park, Jun-Eun; Seok, Mingoo; Jeong, Deog-Kyoon

被引用次数: 2

2024

A 1-TFLOPS/W, 28-nm Deep Neural Network Accelerator featuring Online Compression and Decompression and BF16 Digital In-Memory-Computing Hardware

2024 IEEE CUSTOM INTEGRATED CIRCUITS CONFERENCE, CICC

Zhang, Bo; Moon, Seunghyun; Seok, Mingoo

被引用次数: 1

2024

TICA: Timing Slack Inference and Clock Frequency Adaption Technique for a Deeply Pipelined Near-Threshold-Voltage Bitcoin Mining Core

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Jieyu; He, Weifeng; Zhang, Bo; He, Guanghui; Jin, Jing; Yang, Jun; Seok, Mingoo

被引用次数: 0

2024

Model-Based Study on the Limit of the Dynamic Load Regulation Performance of a Digital Low Dropout Regulator

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Xu, Yichen; Wang, Zhaoqing; Oh, Jonghyun; Seok, Mingoo

被引用次数: 0

2024

STAR-SRAM: 43.06-TFLOPS/W, 1.89-TFLOPS/mm², 400-Kb/mm² Floating-Point SRAM-based Digital Computing-in-Memory Macro in 28-nm CMOS

2024 IEEE CUSTOM INTEGRATED CIRCUITS CONFERENCE, CICC

Lin, Chuan-Tung; Oh, Jonghyun; Lee, Kevin; Seok, Mingoo

被引用次数: 1

2024

AACIM: A 2785-TOPS/W, 161-TOP/mm², <1.17%-RMSE, Analog-In Analog-Out Computing-In-Memory Macro in 28nm

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Wan, Rentao; Lin, Chuan-Tung; Zhang, Bo; Tsvidis, Yannis; Seok, Mingoo

被引用次数: 0

2024

3-D In-Sensor Computing for Real-Time DVS Data Compression: 65-nm Hardware-Algorithm Co-Design

IEEE SOLID-STATE CIRCUITS LETTERS

Nair, Gopikrishnan R.; Nalla, Pragnya S.; Krishnan, Gokul; Anupreetham; Oh, Jonghyun; Hassan, Ahmed; Yeo, Injune; Kasichainula, Kishore; Seok, Mingoo; Seo, Jae-Sun; Cao, Yu

被引用次数: 1

2024

SPADES: A 0.54-GFLOPS/W Sparse Matrix Vector Multiplication Accelerator Featuring On-the-Fly GZIP Decompression for 3.36x Reduction in Off-Chip Data Movement

PROCEEDINGS OF THE 29TH ACM/IEEE INTERNATIONAL SYMPOSIUM ON LOW POWER ELECTRONICS AND DESIGN, ISLPED 2024

Huang, Paul Xuanyuanliang; Tsvidis, Yannis; Seok, Mingoo

被引用次数: 0

2024

93.89% Peak Efficiency 24V-to-1V DC-DC Converter with Fast In-Situ Efficiency Tracking and Power-FET Code Roaming

IEEE 49TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE, ESSCIRC 2023

Wang, Zhaoqing; Li, Mao; Kim, Suhwan; Desai, Nachiket; Krishnamurthy, Ram K.; Lazaro, Orlando; Blanco, Andres; Zhang, Xin; Seok, Mingoo

被引用次数: 0

2023

CDAR-DRAM: Enabling Runtime DRAM Performance and Energy Optimization via In-Situ Charge Detection and Adaptive Data Restoration

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Qin, Yuxuan; Lin, Chuxiong; He, Weifeng; Sun, Yanan; Mao, Zhigang; Seok, Mingoo

被引用次数: 1

2023

A DFT-Compatible In-Situ Timing Error Detection and Correction Structure Featuring Low Area and Test Overhead

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Zhang, Hao; He, Weifeng; Sun, Yanan; Seok, Mingoo

被引用次数: 0

2023

Demo: Experimentation with Wideband Real-Time Adaptive Full-Duplex Radios

PROCEEDINGS OF THE 2023 ACM SIGCOMM 2023 CONFERENCE, SIGCOMM 2023

Levin, Alon S.; Kadota, Igor; Garikapati, Sasank; Zhang, Bo; Jolly, Aditya; Kohli, Manav; Seok, Mingoo; Krishnaswamy, Harish; Zussman, Gil

被引用次数: 0

2023

D6CIM: 60.4-TOPS/W, 1.46-TOPS/mm², 1005-Kb/mm² Digital 6T-SRAM- Based Compute-in-Memory Macro Supporting 1-to-8b Fixed-Point Arithmetic in 28-nm CMOS

IEEE 49TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE, ESSCIRC 2023

Oh, Jonghyun; Lin, Chuan-Tung; Seok, Mingoo

被引用次数: 3

2023

An Output-Capacitor-Free Synthesizable Digital LDO Using CMP-Triggered Oscillator and Droop Detector

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Oh, Jonghyun; Hwang, Young-Ha; Park, Jun-Eun; Seok, Mingoo; Jeong, Deog-Kyoon

被引用次数: 12

2023

60-GHz Single Flux Quantum Pulse Transfer Circuit for Serial Biasing

IEEE TRANSACTIONS ON APPLIED SUPERCONDUCTIVITY

Shukla, Ashish; Kirichenko, Dmitri E. E.; Filippov, Timur V. V.; Sahu, Anubhav; Lehmann, A. Erik; Seok, Mingoo; Gupta, Deepnarayan

被引用次数: 1

2023

PIMCA: A Programmable In-Memory Computing Accelerator for Energy-Efficient DNN Inference

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Bo; Yin, Shihui; Kim, Minkyu; Saikia, Jyotishman; Kwon, Soonwan; Myung, Sungmeen; Kim, Hyunsoo; Kim, Sang Joon; Seo, Jae-Sun; Seok, Mingoo

被引用次数: 21

2023

microASR: 32- μ W Real-Time Automatic Speech Recognition Chip featuring a Bio-Inspired Neuron Model and Digital SRAM-based Compute-In-Memory Hardware

IEEE 49TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE, ESSCIRC 2023

Wang, Dewei; Oh, Jonghyun; Chen, Gregory K.; Knag, Phil; Krishnamurthy, Ram K.; Seok, Mingoo

被引用次数: 0

2023

Improving DNN Hardware Accuracy by In-Memory Computing Noise Injection

IEEE DESIGN & TEST

Cherupally, Sai Kiran; Meng, Jian; Rakin, Adnan Siraj; Yin, Shihui; Seok, Mingoo; Fan, Deliang; Seo, Jae-Sun

被引用次数: 8

2022

Channel Estimation Using Deep Learning on an FPGA for 5G Millimeter-Wave Communication Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chundi, Pavan Kumar; Wang, Xiaodong; Seok, Mingoo

被引用次数: 6

2022

Pulse Interfaces and Current Management Techniques for Serially Biased RSFQ Circuits

IEEE TRANSACTIONS ON APPLIED SUPERCONDUCTIVITY

Shukla, Ashish; Kirichenko, Dmitri E.; Filippov, Timur, V; Sahu, Anubhav; Celik, M. Eren; Seok, Mingoo; Gupta, Deepnarayan

被引用次数: 6

2022

Serial Biasing Technique for Electronic Design Automation in RSFQ Circuits

IEEE TRANSACTIONS ON APPLIED SUPERCONDUCTIVITY

Shukla, Ashish; Filippov, Timur, V; Kirichenko, Dmitri E.; Meher, Sukanya S.; Celik, Mustafa Eren; Seok, Mingoo; Gupta, Deepnarayan

被引用次数: 2

2022

MEDAC: A Metastability Condition Detection and Correction Technique for a Near-Threshold-Voltage Multi-Voltage-/Frequency-Domain Network-on-Chip

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lin, Chuxiong; He, Weifeng; Sun, Yanan; Pei, Bingxi; Chundi, Pavan Kumar; Mao, Zhigang; Seok, Mingoo

被引用次数: 5

2021

An Area-Efficient Scannable In Situ Timing Error Detection Technique Featuring Low Test Overhead for Resilient Circuits

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Zhang, Hao; He, Weifeng; Sun, Yanan; Seok, Mingoo

被引用次数: 4

2021

Nanowatt Acoustic Inference Sensing Exploiting Nonlinear Analog Feature Extraction

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Minhao; Liu, Hongjie; Shan, Weiwei; Zhang, Jun; Kiselev, Ilya; Kim, Sang Joon; Enz, Christian; Seok, Mingoo

被引用次数: 9

2021

Always-On Sub-Microwatt Spiking Neural Network Based on Spike-Driven Clock- and Power-Gating for an Ultra-Low-Power Intelligent Device

FRONTIERS IN NEUROSCIENCE

Chundi, Pavan Kumar; Wang, Dewei; Kim, Sung Justin; Yang, Minhao; Cerqueira, Joao Pedro; Kang, Joonsung; Jung, Seungchul; Kim, Sangjoon; Seok, Mingoo

被引用次数: 4

2021

CDAR-DRAM: An In-situ Charge Detection and Adaptive Data Restoration DRAM Architecture for Performance and Energy Efficiency Improvement

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Lin, Chuxiong; He, Weifeng; Sun, Yanan; Mao, Zhigang; Seok, Mingoo

被引用次数: 4

2021

Leveraging Noise and Aggressive Quantization of In-Memory Computing for Robust DNN Hardware Against Adversarial Input and Weight Attacks

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Cherupally, Sai Kiran; Rakin, Adnan Siraj; Yin, Shihui; Seok, Mingoo; Fan, Deliang; Seo, Jae-sun	2021
被引用次数: 8	
FLEET-Fast Lanes for Expedited Execution at 10 Terabits: Program Overview	
IEEE INTERNET COMPUTING	
Douglis, Fred; Robertson, Seth; van den Berg, Eric; Micallef, Josephine; Pucci, Marc; Aiken, Alex; Bergman, Keren; Hattink, Maarten; Seok, Mingoo	
被引用次数: 5	2021
Modeling and Optimization of SRAM-based In-Memory Computing Hardware Design	
PROCEEDINGS OF THE 2021 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION (DATE 2021)	
Saikia, Jyotishman; Yin, Shihui; Cherupally, Sai Kiran; Zhang, Bo; Meng, Jian; Seok, Mingoo; Seo, Jae-Sun	
被引用次数: 8	2021
CDAR-DRAM: An In-situ Charge Detection and Adaptive Data Restoration DRAM Architecture for Performance and Energy Efficiency Improvement	
Design Automation Conference	
Chuxiong Lin; Weifeng He; Yanan Sun; Zhigang Mao; Mingoo Seok	
被引用次数: 0	2021
0.5-1-V, 90-400-mA, Modular, Distributed, 3 x 3 Digital LDOs Based on Event-Driven Control and Domino Sampling and Regulation	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Kim, Sung Justin; Kim, Dongkwun; Pu, Yu; Shi, Chunlei; Chang, Soo Bong; Seok, Mingoo	
被引用次数: 15	2021
Leveraging Noise and Aggressive Quantization of In-Memory Computing for Robust DNN Hardware Against Adversarial Input and Weight Attacks	
Design Automation Conference	
Sai Kiran Cherupally; Adnan Siraj Rakin; Shihui Yin; Mingoo Seok; Deliang Fan; Jae-sun Seo	
被引用次数: 0	2021
Introduction to the Special Issue on the 2019 IEEE International Solid-State Circuits Conference (ISSCC)	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Carusone, Tony Chan; Seok, Mingoo; Chang, Hsie-Chia; Chang, Meng-Fan	
被引用次数: 0	2020
XNOR-SRAM: In-Memory Computing SRAM Macro for Binary/Ternary Deep Neural Networks	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Yin, Shihui; Jiang, Zhewei; Seo, Jae-Sun; Seok, Mingoo	
被引用次数: 289	2020
C3SRAM: An In-Memory-Computing SRAM Macro Based on Robust Capacitive Coupling Computing Mechanism	

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiang, Zhewei; Yin, Shihui; Seo, Jae-Sun; Seok, Mingoo

被引用次数: 206

2020

Catena: A Near-Threshold, Sub-0.4-mW, 16-Core Programmable Spatial Array Accelerator for the Ultralow-Power Mobile and Embedded Internet of Things

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Cerqueira, Joao P.; Repetti, Thomas J.; Pu, Yu; Priyadarshi, Shivam; Kim, Martha A.; Seok, Mingoo

被引用次数: 13

2020

A Bi-Directional, Zero-Latency Adaptive Clocking Circuit in a 28-nm Wide AVFS System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Shan, Weiwei; Dai, Wentao; Wan, Liang; Lu, Minyi; Shi, Longxing; Seok, Mingoo; Yang, Jun

被引用次数: 15

2020

Vesti: Energy-Efficient In-Memory Computing Accelerator for Deep Neural Networks

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yin, Shihui; Jiang, Zhewei; Kim, Minkyu; Gupta, Tushar; Seok, Mingoo; Seo, Jae-Sun

被引用次数: 42

2020

Recursive Synaptic Bit Reuse: An Efficient Way to Increase Memory Capacity in Associative Memory

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Guan, Tianchan; Zeng, Xiaoyang; Seok, Mingoo

被引用次数: 1

2019

A Near-Threshold Spiking Neural Network Accelerator With a Body-Swapping-Based In Situ Error Detection and Correction Technique

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Seongjong; Cerqueira, Joao Pedro; Seok, Mingoo

被引用次数: 14

2019

Recursive Binary Neural Network Training Model for Efficient Usage of On-Chip Memory

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Guan, Tianchan; Liu, Peiye; Zeng, Xiaoyang; Kim, Martha; Seok, Mingoo

被引用次数: 4

2019

Design of an Always-On Deep Neural Network-Based 1- μ W Voice Activity Detector Aided With a Customized Software Model for Analog Feature Extraction

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Minhao; Yeh, Chung-Heng; Zhou, Yiyin; Cerqueira, Joao P.; Lazar, Aurel A.; Seok, Mingoo

被引用次数: 64

2019

Energy-Efficient Hybrid Analog/Digital Approximate Computation in Continuous Time (vol 51, pg 1514, 2016)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Guo, Ning; Huang, Yipeng; Mai, Tao; Patil, Sarvil; Cao, Chi; Seok, Mingoo; Sethumadhavan, Simha; Tsividis, Yannis

被引用次数: 1

2018

An Area-Efficient Microprocessor-Based SoC With an Instruction-Cache Transformable to an Ambient Temperature Sensor and a Physically Unclonable Function

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Jiangyi; Yang, Teng; Yang, Minhao; Kinget, Peter R.; Seok, Mingoo

被引用次数: 22

2018

In Situ and In-Field Technique for Monitoring and Decelerating NBTI in 6T-SRAM Register Files

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yang, Teng; Kim, Doyun; Li, Jiangyi; Kinget, Peter R.; Seok, Mingoo

被引用次数: 7

2018

In Situ Error Detection Techniques in Ultralow Voltage Pipelines: Analysis and Optimizations

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Jin, Wei; Kim, Seongjong; He, Weifeng; Mao, Zhigang; Seok, Mingoo

被引用次数: 8

2017

Near- and Sub-Vt Pipelines Based on Wide-Pulsed-Latch Design Techniques

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jin, Wei; Kim, Seongjong; He, Weifeng; Mao, Zhigang; Seok, Mingoo

被引用次数: 8

2017

ANALOG COMPUTING IN A MODERN CONTEXT: A LINEAR ALGEBRA ACCELERATOR CASE STUDY

IEEE MICRO

Huang, Yipeng; Guo, Ning; Seok, Mingoo; Tsividis, Yannis; Sethumadhavan, Simha

被引用次数: 29

2017

A Fully Integrated Digital Low-Dropout Regulator Based on Event-Driven Explicit Time-Coding Architecture

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Doyun; Seok, Mingoo

被引用次数: 57

2017

Triple-Mode, Hybrid-Storage, Energy Harvesting Power Management Unit: Achieving High Efficiency Against Harvesting and Load Power Variabilities

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Jiangyi; Seo, Jae-sun; Kymissis, Ioannis; Seok, Mingoo

被引用次数: 32

2017

Temporarily Fine-Grained Sleep Technique for Near- and Subthreshold Parallel Architectures

Ultra-Compact and Robust Physically Unclonable Function Based on Voltage-Compensated Proportional-to-Absolute-Temperature Voltage Generators

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Jiangyi; Seok, Mingoo

被引用次数: 73

2016

High-Accuracy Compressed Sensing Decoder Based on Adaptive (l0, l1) Complex Approximate Message Passing: Cross-layer Design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zheng, Le; Wu, Zhenzhi; Seok, Mingoo; Wang, Xiaodong; Liu, Quanhua

被引用次数: 14

2016

Energy-Efficient Neuromorphic Classifiers

NEURAL COMPUTATION

Marti, Daniel; Rigotti, Mattia; Seok, Mingoo; Fusi, Stefano

被引用次数: 22

2016

Energy-Efficient Hybrid Analog/Digital Approximate Computation in Continuous Time

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Guo, Ning; Huang, Yipeng; Mai, Tao; Patil, Sharvil; Cao, Chi; Seok, Mingoo; Sethumadhavan, Simha; Tsividis, Yannis

被引用次数: 67

2016

Compact and Supply-Voltage-Scalable Temperature Sensors for Dense On-Chip Thermal Monitoring

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yang, Teng; Kim, Seongjong; Kinget, Peter R.; Seok, Mingoo

被引用次数: 47

2015

Variation-Tolerant, Ultra-Low-Voltage Microprocessor With a Low-Overhead, Within-a-Cycle In-Situ Timing-Error Detection and Correction Technique

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Seongjong; Seok, Mingoo

被引用次数: 91

2015

Circuits for a Cubic-Millimeter Energy-Autonomous Wireless Intraocular Pressure Monitor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ghaed, Mohammad Hassan; Chen, Gregory; Razi-ul Haque; Wieckowski, Michael; Kim, Yejoong; Kim, Gyouho; Lee, Yoonmyung; Lee, Inhee; Fick, David; Kim, Daeyeon; Seok, Mingoo; Wise, Kensall D.; Blaauw, David; Sylvester, Dennis

被引用次数: 102

2013

A Millimeter-Scale Energy-Autonomous Sensor System With Stacked Battery and Solar Cells

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fojtik, Matthew; Kim, Daeyeon; Chen, Gregory; Lin, Yu-Shiang; Fick, David; Park, Junsun; Seok, Mingoo; Chen, Mao-Ter; Foo, Zhiyoong; Blaauw, David; Sylvester, Dennis

被引用次数: 94

2013

Achieving Ultralow Standby Power With an Efficient SCCMOS Bias Generator

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Yoonmyung; Seok, Mingoo; Hanson, Scott; Sylvester, Dennis; Blaauw, David

被引用次数: 5

2013

Design Methodology for Voltage-Overscaled Ultra-Low-Power Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jeon, Dongsuk; Seok, Mingoo; Zhang, Zhengya; Blaauw, David; Sylvester, Dennis

被引用次数: 22

2012

A Super-Pipelined Energy Efficient Subthreshold 240 MS/s FFT Core in 65 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeon, Dongsuk; Seok, Mingoo; Chakrabarti, Chaitali; Blaauw, David; Sylvester, Dennis

被引用次数: 66

2012

A Portable 2-Transistor Picowatt Temperature-Compensated Voltage Reference Operating at 0.5 V

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Seok, Mingoo; Kim, Gyouho; Blaauw, David; Sylvester, Dennis

被引用次数: 349

2012

CAS-FEST 2010: Mitigating Variability in Near-Threshold Computing

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Seok, Mingoo; Chen, Gregory; Hanson, Scott; Wieckowski, Michael; Blaauw, David; Sylvester, Dennis

被引用次数: 44

2011

Robust Clock Network Design Methodology for Ultra-Low Voltage Operations

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Seok, Mingoo; Blaauw, David; Sylvester, Dennis

被引用次数: 15

2011

A Low-Voltage Processor for Sensing Applications With Picowatt Standby Mode

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hanson, Scott; Seok, Mingoo; Lin, Yu-Shiang; Foo, ZhiYoong; Kim, Daeyeon; Lee, Yoonmyung; Liu, Nurrachman; Sylvester, Dennis; Blaauw, David

被引用次数: 132

2009

Nanometer device scaling in subthreshold logic and SRAM

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hanson, Scott; Seok, Mingoo; Sylvester, Dennis; Blaauw, David

被引用次数: 112

2008

Exploring variability and performance in a sub-200-mV processor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

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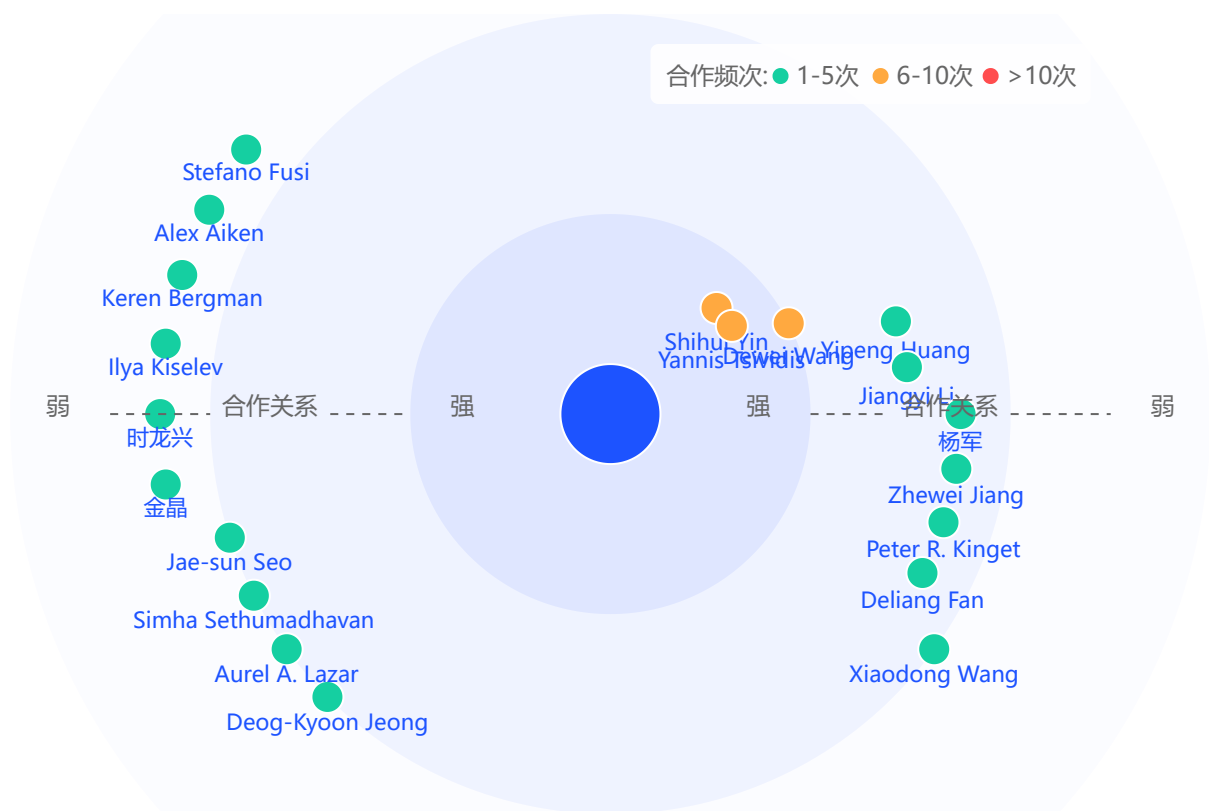
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